RAJIV GANDHI COLLEGE OF ENGINEERING AND TECHNOLOGY

COMPUTER SCIENCE AND ENGINEERING

ASSIGNMENT - 2

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Subject Name : Web Technology

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Unit : 2

Teacher's Sign :

Department Vision And Mission

Vision

Cultivate student as technocracts, researchers. and entrepreneurs in the field of computer science and engineering.

Mission.

- To impart quality education in Computer Science and Engineering by means of learning techniques and value -added courses.
- To inculcate professional excellence and research culture by encourage projects in cutting -edge technologies, through industry interactions.
- To build leadership skills, ethical values and teamwork among the students.
- To strengthen the collaboration of department and industry through internships, mentorships and professional body activities.

PART-A

1) Define Literals.

* Literals are fixed values that are used in programming languages, such as HTML, CSS, Javascript and others.

* Literals are values that are written exactly as they are meant to be intrepreted without any additional processing or evaluations.
  - → Numbers (eg: 123)
  - → Text strings (eg: Hello)
  - → Boolean values (eg: true or False)
  - → Null values (eg: Null)

2) Define cookies:

* Cookies are small text files stored on a user's device by a web browser, containing data exchanged between a website and the user's browser

3) Differentiate between servlet and applet?

Servlet	Applet
* Runs on the server side	* Runs on the client side
* Excutes on the web server	* Executes on the user's web browser.
* Handles HTTP requests and responses	* Embedded in an HTML page.

4) List the life cycle of servlet?

- Loading
- Instantiation
- Initialization
- Service
- Destruction.

5) Compare servlet with CGI?

1. <u>Performance</u>: Servlets are faster than CGI because they are executed within the web server as a single process with multiple threads, whereas CGI creates a new process for each requests, leading to higher overhead.

2. <u>Memory usage</u>: Servlet uses memory compares to CGI since they share resources within the same JVM while CGI spawns separate processes.

<u>PART - B</u>

1) Explain the components of JSP in detail.

<u>JSP Syntax</u>: The syntax of JSP is almost similar to that of XML. The general rules are,

1. Tags must have their matching end tags.
2. Attributes must appear in the start tags.
3. Attribute values in the tag must be quoted

White space within the body text of JSP page are preserved during the translation phase - to use special character such as ' \ ' and

and a '/' character before it.

JSP components

A JSP page consists of the following components.

1) Directives
2) Scripting elements.
   - → Expressions
   - → Declarations
   - → Scriplets.
3) Standard Actions
4) Custom tags

JSP Tags

→ <% @ ..... %> - JSP directives such as page and includes.

→ <% = ... %> → JSP expressions

→ < % ... %> JSP scripts that can contain arbitary Java statements.

1) Directives :

* A directives element is a JSP page provide global information amount a particular JSP page.
+ The syntax of JSP directive is

<%, @ directive-name alternative. %>

The three standard directive available in all JSP environment.

1. Page
2. include
3. Taglib.

* Page directive

The page directive defines attributes that notify the web container about the general settings of a JSP page

SYNTAX: <% @ page attribute = "value" attribute = "value" %>

* Include directive:

Include directive are two used to specify the name of the files to be inserted during the compilation of the JSP page. This directive is to include the HTML JSP or service file into a JSP servlet. This a static inclusion to a file

Syntax: <%. @ include file = " filename " %>

2) Spring elements:

The 3 different type of scripting elements are:

1) Expressions
2) Declarations
3) Scriplets.

Expression:

→ Syntax of JSP expression is <% = expression %>

→ JSP expression starts with <% = and ends with %>

→ The result of expression is connected to the string to get displayed.

Ex: <%= "Hello World! " %>

* declarations.

→ The syntax of JSP declarations is
<% ! // declare all the variable here %>

→ JSP declaration begins with <% ! ends with %>

* Scriplets

The syntax of JSP scriplets.

<%. // Java code %>

Eg: <% string username = use;
username = request.get parameter ("user name");
%>

variables available to the JSP scriplets are

1) Request - HTTP servlet Request
2) Response - HTTP servlet Response
3) Session - HTTP session request
4) out - is an object of output streams.

3) Custom tags

The taglib directive makes custom actions available in the current page through the use of a tag library.

The syntax of the directive is

<%. @ taglibrier = "tag Library URI" prefix = 'tag preffic '%>

Attribute	Value
Taglibaray URI	- The URI of a tag library descriptive
Tag prefix	- A unique prefix used to identify custom tags are used later in the page.

2) Explain how will you Acces a database from JSP ?

* All the web application usually interact with difficult types of databases. Databases are basically used to store various types of information for difficult purposes.

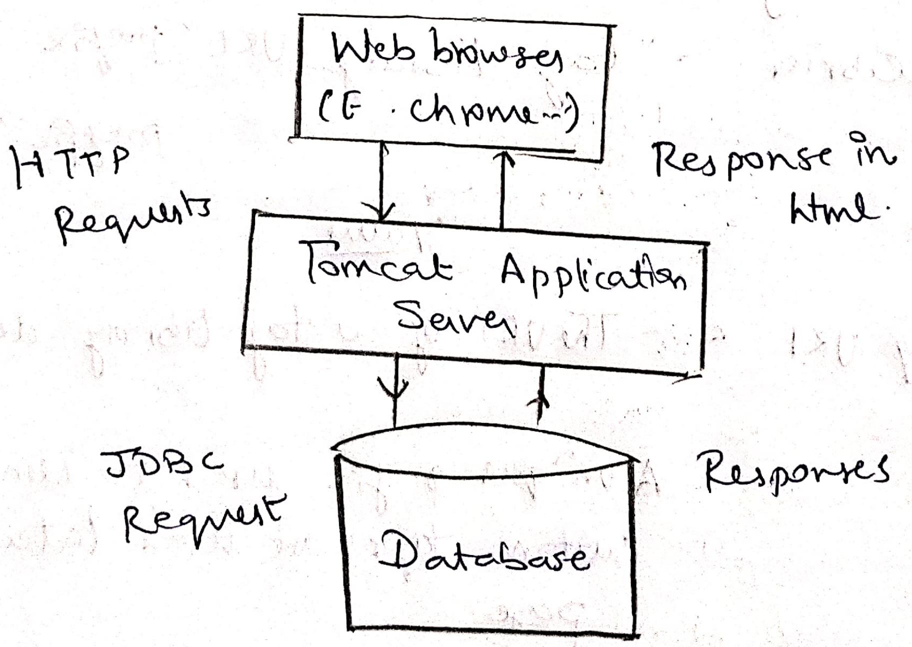
* JSP pages are look links like regular html page with the inclusion of dynamic behaviour.

* JSP pages allow the develop to inteact with database by using JDBC code.

* By using scriplets we can include database programming (JDBC) code.

* For databases connectivity we write the following code in JSP page.

The architecture of accessing a database by using JSP.



Open connectivity code:

<%.@ page language = "Java" import = "Java.sql "%>

<% // load driver class file of oracle

```
class.for.Name ("oracle.jdbc.driver.oracledriver");
Connection con = (Driver Manager.get Connection)"
                                    (jdbc:
mysql://localhost:3306/Testdb.root." root")
%>
```

Statement code:

```
<%  Statement stmt = conn.create statement();
    string query = "select * from students"
   stmt.execute Query (query);
 %>
```

Class connectivity code:

```
<%  Stmt.close();
conn.close()
 %>
```

Eg:

```
<html>
 <head>
 <title> Registration </title>
 </head>
<body>
<form action="reg.jsp" method="post">
Email : <input type ="text" name ="email" />
 <br>
Last name : <input type =" text name ="userid" />
                                          <br>
Password : <input type =" password" name =" pass "/>
                                          <br>
<input type =" submit "/>
 </form>
 </body>
 </html>
```

output

Email : xyz@xyz.com

FirstName: xyz

LastName: abc

Username: abcxyz

password: ......

Submit

Registration successfull.